# Tutorial No : 10

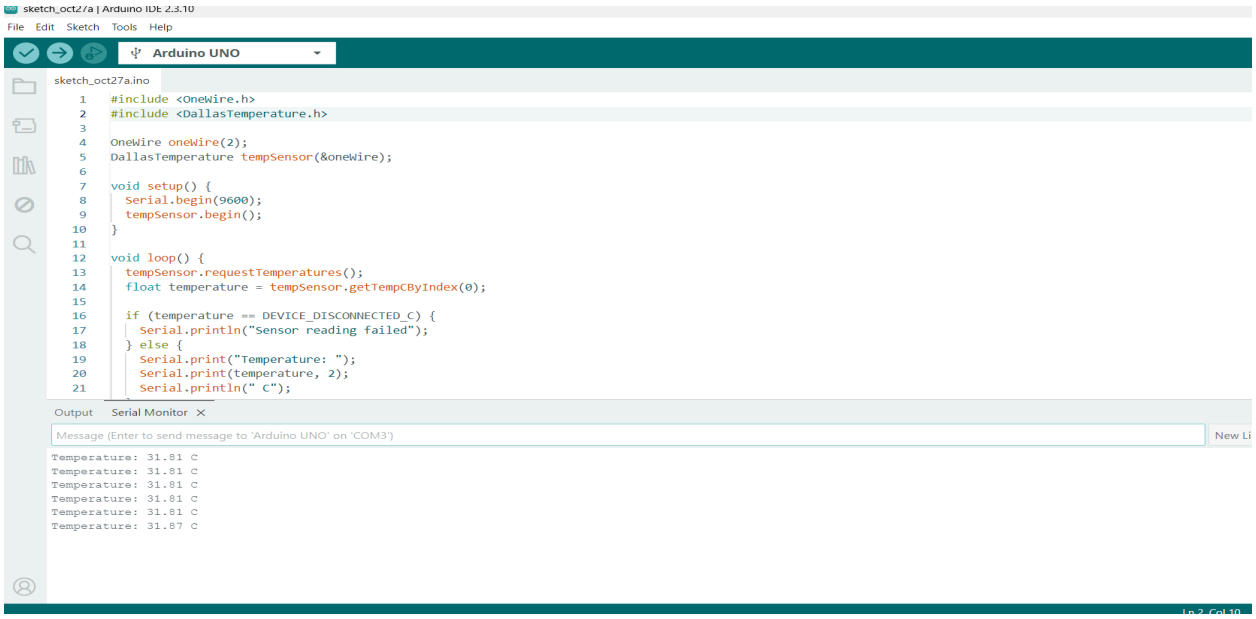
**Tutorial Title:** Implement UART communication using Data Structures in C
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**Class:** SY **Div:** 2 **PRN No:** B25ET1099 **Batch:** B2

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**Program:**

```
#include <OneWire.h>
#include <DallasTemperature.h>
OneWire oneWire(2);
DallasTemperature tempSensor(&oneWire);
void setup() {
  Serial.begin(9600);
  tempSensor.begin();
}
void loop() {
  tempSensor.requestTemperatures();
  float temperature = tempSensor.getTempCByIndex(0);
 if (temperature == DEVICE_DISCONNECTED_C) {
    Serial.println("Sensor reading failed");
  } else {
    Serial.print("Temperature: ");
    Serial.print(temperature, 2);
    Serial.println(" C");   }
  delay(2000);  }
```

**Output:**


sketch_oct27a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_oct27a.ino
#include <OneWire.h>
#include <DallasTemperature.h>
OneWire oneWire(2);
DallasTemperature tempSensor(&oneWire);
void setup() {
Serial.begin(9600);
tempSensor.begin();
}
void loop() {
tempSensor.requestTemperatures();
float temperature = tempSensor.getTempCByIndex(0);
if (temperature == DEVICE_DISCONNECTED_C) {
Serial.println("Sensor reading failed");
} else {
Serial.print("Temperature: ");
Serial.print(temperature, 2);
Serial.println(" C");
Output
Serial Monitor
Message (Enter to send message to 'Arduino UNO' on 'COM3')
Temperature: 31.81 C
Temperature: 31.81 C
Temperature: 31.81 C
Temperature: 31.81 C
Temperature: 31.81 C
Temperature: 31.87 C